

Using functional diversity as an indicator of restoration success of a cut-over bog

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ABSTRACT

A trait-based approach was used to evaluate the restoration success of a moss layer transfer technique on the herbaceous community of a cut-over bog. In order to understand the herbaceous dynamics following restoration, the bog was monitored prior to restoration and biannually over ten subsequent years, in two initial drainage conditions (dry peat fields and wet ditches), and compared to seven undisturbed bogs. Herbaceous community found prior to and in the early stage after restoration were characterized by species associated to mesic soil conditions with a fen-like pH, ruderal or wetland species, species with mycorrhiza or exotic species. These species did not persist in the latter stage after restoration and were replaced by species that are carnivorous, are associated to variable water availability and show mostly sexual reproduction. The extent of changes in traits composition of the herbaceous community between the early and latter stages after restoration showed the importance of multiple-year monitoring to evaluate restoration projects. The trait composition found in the latter stage after restoration appeared to converge toward the reference ecosystem where the community was characterized by species with large seeds, associated to humid soil conditions and a forest or peatland habitat. Lower trait diversity was found in the latter stage after restoration than in the reference ecosystem, especially in peat fields. In addition, species and functional diversity pattern varied with the initial drainage conditions, decreasing in peat fields while remaining stable in ditches. This suggests that different initial topography for restoration might help increase species and functional diversity of a managed ecosystem. Finally, our results outline measures to improve the moss layer transfer technique: sowing herbaceous species with large seeds in the latter stage after restoration, applying the restoration technique as soon as exploitation has terminated or removing all vegetation (including seeds and rhizomes) from the site to be restored.

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1. Introduction

Ecological restoration can draw upon a wide array of techniques, yet assessment of success is an essential component of any restoration strategy. In that respect, the need for standardized evaluation methods and multi-year monitoring are critical (Bernhardt et al., 2005; SERIS & PWG, 2004). When restoring an entire ecosystem (Ehrenfeld, 2000), practitioners usually improve abiotic conditions and/or translocate propagules in order to overcome environmental filters (e.g. inappropriate light or water conditions, nutrient limitation, competition, etc.) that limit the establishment and maintenance of species typical of the undisturbed state.

However, unforeseen and undesirable species can also benefit from the environmental changes following restoration and may compete with species targeted by the restoration project (Young et al., 2005). Indeed, in the first years following restoration, the ecosystem is highly dynamic (Sarr, 2002) and rapid shifts in communities can be observed (Kellogg and Bridgham, 2002). Multiple years of monitoring are thus required to accurately assess restoration success (Verhagen et al., 2001).

The use of a reference ecosystem is encouraged to set restoration objectives and to evaluate success. It is usually composed of a set of sites representing the ecological variability of an undisturbed ecosystem within a specific region (SERIS & PWG, 2004; Landres et al., 1999; White and Walker, 1997). Ecological variability can be measured using biotic community attributes such as species composition and diversity. Restoration success can then be evaluated in a trajectory analysis, where the impact of restoration is assessed through time by evaluating the distance between the restored and reference ecological attributes (Young et al., 2005). In case

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of a successful intervention, species assemblages of the restored ecosystem will become closer to those found in the reference ecosystem (Dufour and Piégay, 2009; Verhagen et al., 2001).

One promising approach for evaluating restoration success is based on species traits (Sandel et al., 2011; Tullios et al., 2009; Gondard et al., 2003; Pywell et al., 2003). Traits have been shown to be useful in characterizing community development over an extended timeframe or a wide variety of conditions (e.g. Laliberté et al., 2010; Lacourse, 2009). A trait-based approach can help clarify the community assembly dynamic since the potential of a species to establish or persist under a given set of environmental filters is largely determined by its traits (Lavorel et al., 2007). Traits known to be associated to an ecosystem process critical for restoration success could guide species selection (Sandel et al., 2011). For instance, Pywell et al. (2003) suggested that stress-tolerant species be introduced several years after restoration, when the community of a grassland ecosystem has begun to stabilize. A trait approach has been used to measure restoration success in ecosystems such as rivers (Tullios et al., 2009), grasslands (Pywell et al., 2003), pine forests and steppes (Gondard et al., 2003). However, these previously published studies rely mostly on data covering less than five years' time. As restoration projects have rarely been monitored on a longer term (Follstad Shah et al., 2007), opportunities for using a trait approach for multi-year assessment of restoration success have been limited (Kahmen et al., 2002).

Worldwide, peatlands have been exploited for fuel or drained for forestry and agriculture for centuries (Jutras et al., 2006; Renou et al., 2006; Chapman et al., 2003; Dachnowski-Stokes, 1934) and their restoration has been a topic of discussion for more than 60 years (Dachnowski-Stokes, 1939). Peatlands provide beneficial services to humans, such as nutrient cycling, carbon storage, biodiversity support and water regulation, which make their restoration critical (Kimmel and Mander, 2010). Over the last two decades, specific restoration techniques have been developed for ombrotrophic peatlands (bogs) subjected to industrial peat exploitation through vacuum suction, a technique that became common in the 1960s. These so called cut-over bogs lose their carbon accumulating function (Thormann et al., 1999a), as keystone *Sphagnum* species cannot re-establish spontaneously on the dry peatland fields that remain after harvesting (hereafter called "peat fields"; Poulin et al., 2005). Cut-over peatlands are devoid of any seed banks (Salonen, 1987), have a low as well as highly variable water level due to abundant ditches (Price et al., 2003), have a low concentration of phosphorus (0.26 mg/g for total P and under detection threshold for PO_4 ; Andersen et al., 2011) and have unstable surfaces created by frost heaving (Groeneveld and Rochefort, 2002). In North America, research on peatland restoration resulted in an ecosystem-scale technique called "moss layer transfer", which consists of collecting a thin layer of vegetation from nearby undisturbed bog and spreading plant fragments over an area of cut-over peatland ten times larger (Rochefort et al., 2003). For ombrotrophic peatlands, the primary purpose of this technique is the re-establishment of *Sphagnum* cover and successful rewetting of the site in order to restore the carbon accumulation processes (Gorham and Rochefort, 2003). Typical bog plant diversity can also be expected to re-establish as all types of plant propagules (rhizomes, roots, seeds) are transferred along with moss fragments (Rochefort, 2000).

A nine-year monitoring of a bog restored using the moss layer transfer technique has shown that mosses and herbaceous species are the most successful at establishing after restoration (Rochefort et al., unpublished results). In undisturbed bogs though, the herbaceous community typically represents a minor component compared to mosses or shrubs (Moore et al., 2002). Consequently, the establishment success of herbs in restored bog merits more detailed investigation. Our objective was to understand herbaceous

dynamics in a restored bog in comparison to a reference ecosystem. To do so, we identified traits associated with the presence of herbaceous species at different stand stages (prior to and up to ten years after restoration) for two initial drainage conditions (dry peat fields and wet ditches) in a restored bog as well as in a reference ecosystem (see Table 1 for hypotheses). Then, we (1) evaluated whether the traits of the restored bog converged toward the reference ecosystem to assess restoration success, and we (2) estimated species, functional and trait diversity through time to better characterize the restored bog.

2. Materials and methods

2.1. Study site

The experimental site is located within the Bois-des-Bel undisturbed bog (47°48'N, 69°31'W), which covers a total of 189 ha. It is located on the south shore of the St. Lawrence River near Rivière-du-Loup (Québec, Canada). In this region, annual average temperature is 3.2 °C (Meteorological Service of Canada, 2011). The region receives 963 mm of precipitation, of which 685 mm is rainfall (Meteorological Service of Canada, 2011). Peatlands (mainly bogs) represent 8% of the regional landscape (Pellerin, 2003), which is covered mostly by agricultural fields (~66%; Robitaille and Saucier, 1998) and to a lesser extent by woodlots (15%; Lachance et al., 2005). Peatlands in this region are becoming increasingly forested; their tree cover has increased in the range of 23–57% between 1948 and 1995 due to a combination of factors including fire and drier than average climate (Pellerin and Lavoie, 2003). In this region, open peatlands have higher species richness and different herbaceous species assemblages (e.g. *Carex pauciflora* Lightf., *Drosera rotundifolia* L., *Eriophorum vaginatum* L., *Maianthemum trifolium* (L.) Sloboda, *Rubus chamaemorus* L.) than forested peatlands (e.g. *Carex trisperma* Dewey; Lachance et al., 2005; Gignac et al., 2004; Poulin et al., 2002; Pellerin and Lavoie, 2000). The reference ecosystem was composed of seven bogs that were: (1) at least 100 m long, (2) more than 75% moss-covered, (3) located within 80 km from the restored bog and (4) represented the last and least disturbed bogs of the region.

The experimental site was mechanically exploited for peat between 1970 and 1980. In 1999–2000, a contiguous section of 8.4 ha consisting of eight 30 m wide peat fields separated by eight 1.5 m wide ditches was restored using the moss layer transfer technique. The study design was a quasi-experiment (sensu Manly, 1992; Eberhardt and Thomas, 1991), as peatland restoration implies raising the water table level thus impeding complete randomization within the site (Block et al., 2001).

2.2. Steps of the moss layer transfer technique

The first step in the application of the moss layer transfer technique was the flattening and leveling of the peat surface. To do so, vegetation and peat were scraped off the peat fields and used later to block ditches. The top five to ten centimeters of the moss layer of an ombrotrophic peatland located 20 km south of the experimental site (47°47'N, 69°29'W) was collected. This site was dominated by *Sphagnum rubellum* Wils. ($39.8 \pm 27.3\%$; Rochefort et al., 2002) with a sparse cover of herbaceous species, mainly composed of *M. trifolium* ($1.1 \pm 2.3\%$), *Sarracenia purpurea* L. ($0.7 \pm 1.2\%$) and *R. chamaemorus* ($0.5 \pm 0.8\%$; D'Astous, 2012). Plant fragments (diaspores of *Sphagnum* and mosses, seeds, roots and rhizomes of vascular plants) were transferred to the restoration site in a ratio of 1:10 (one ha of vegetation material spread over ten ha of the restored site; Quinty and Rochefort, 2003). Straw mulch was added

Table 1

Plant traits description and literature review of traits associated negatively (–) or positively (+) with the reference ecosystem (undisturbed bogs) or with cut-over bogs (previously exploited by vacuum suction). Four traits associated to either plant survival (PS) or reproductive success (RS) and four traits showing optimum species distribution in relation to environmental gradients (EG) were used. The traits have been inflated into values ranging between 0 and 1 according to their gradient (see bullet points a–c for description).

Traits	Process	States	Code	Inflated value	Description	Reference ecosystem	Cut-over bogs
Nutritional adaptations ^a	PS	Carnivorous Absence	Carnivorous	1	Carnivorous	+ ^g	+ ^m
				0	Not carnivorous	– ^g	– ^m
Nutrient uptake ^a	PS	Presence Absence	Mycorrhiza	1	Presence of any kind of mycorrhiza	– ^h	+ ⁿ
				0	Absence of any kind of mycorrhiza		
Reproduction type ^c	RS	Sexual	Sexual	0	Mostly asexual reproduction	+ ⁱ	– ^o
				0.5	Mostly sexual reproduction	– ⁱ	+ ^o
				1	Only sexual reproduction	– ⁱ	+ ^o
Seed size ^c	RS	Length	Seed	0	<0.1 mm; very small	– ^j	+ ^p
				0.25	0.1–1.99 mm; small	– ^j	+ ^p
				0.5	2–2.99 mm; medium	– ^j	+ ^p
				0.75	3–4.99 mm; large	+ ^j	– ^p
				1	5–10 mm; very large	+ ^j	– ^p
Acidity tolerance ^a	EG	Fen Bog	High.pH	1	pH > 5.7	– ^k	
				0	pH ≤ 5.7	+ ^k	– ⁿ
Habitat ^b	EG	Forest	HabFor	1	Species found in forest	– ^g	– ^q
		Peatland ^e	HabPea	1	Species mainly confined to bog or fens	+ ^g	– ^q
		Ruderal	HabRud	1	Species colonizing disturbed areas like road side	– ^g	+ ^q
		Wetland ^f	HabWet	1	Species that generally prefer wetland	– ^g	+ ^q
Status in Québec ^a	EG	Introduced Indigenous	Exotic	1	Exotic	– ^l	
				0	Native	+ ^l	
Water preference ^{b,d}	EG	Generalist	WatGen	1	Soil goes from xeric to wet conditions (variable availability)	– ^g	+ ^r
		Mesic	WatMes	1	Soil is moist for part of the growing season	– ^g	+ ^r
		Humid	WatHum	1	Soil wet in the growing season (i.e. water table close to surface)	+ ^g	– ^r

^a Only one state was used in the analyses in order to avoid collinearity.

^b Nominal traits with more than two states were inflated to 1 and analyzed separately as dummy variables. This makes a total of thirteen trait states as indicated by the column 'code'.

^c Ordinal traits were assigned a value between 0 and 1.

^d B.C. Ministry of Forests and B.C. Ministry of Environment, Lands and Parks, 1998; ^e Garneau, 2001; ^f MDDEP, 2008; ^g Payette and Rochefort, 2001; ^h Thormann et al., 1999b; ⁱ Pywell et al., 2003; ^j Turnbull et al., 1999; ^k Gorham and Janssens, 1992; ^l Lavoie and Rochefort, 1996; ^m Karagatzides and Ellison, 2009; ⁿ Andersen et al., 2006; ^o Campbell et al., 2003; ^p Campbell and Rochefort, 2003; ^q Poulin et al., 2005; ^r Shantz and Price, 2006.

to improve micro-climatic conditions and protect plant fragments from desiccation (Quinty and Rochefort, 2003). Drainage ditches were blocked every 50 m with highly compacted peat while bunds were built with previously leveled peat in order to distribute water homogeneously over the site (Shantz and Price, 2006). A 150 kg/ha dose of phosphorus from McInnes granular phosphate rock (0–13–0) was applied prior to the first growing season to favor the establishment of *Polytrichum* to diminish frost heaving (Groeneveld and Rochefort, 2005).

2.3. Data collection

Systematic monitoring was set up to evaluate species composition in peat fields and ditches prior to restoration as well as one, three, five, seven and ten years after restoration (year one was omitted for ditches). Equidistant transects were set up over peat fields and ditches (respectively ten and six points per transect for a total of 4400 and 1475 survey points over the entire experimental site). At each point, all species touching a rod of two centimeters in diameter were recorded. The tenth year, about half of the points were surveyed, as this quantity was extensive enough to capture most species diversity. The seven undisturbed bogs composing the reference ecosystem were surveyed in 2007. Ground cover (%) of herbaceous species was visually estimated in ten one m² equidistant quadrats distributed along a linear transect crossing each site. One of the bogs used as a reference ecosystem, the undisturbed section of Bois-des-Bel, was surveyed again in 2009 with a greater

number of quadrats (D'Astous, 2012). Ground cover of herbaceous species (%) was evaluated in six quadrats of 70 cm diameter for each of the larger fifteen plots (3 m × 8 m). A third of those plots were located in open sections (less than 50% cover of trees higher than 1.5 m) of the bog. The few specimens that could only be identified to the genus level due to the absence of distinguishable characters were removed from the dataset, but their occurrences were extremely low (<1%). Nomenclature follows the Integrated Taxonomic Information System (ITIS; Government of Canada, 2009). A list of all herbaceous species found is shown in supplementary material (see the 94 species in Appendix A, Table 3).

2.4. Species ecological traits

Information on the species traits was obtained from the TOPIC database (Traits of Plants in Canada; see <http://topic.nrcan.gc.ca/>). A total of eight categorical traits were selected as they illustrated different ecological processes: four traits were associated to either plant survival or reproductive success and four traits were related to optimum species distribution along environmental gradients (i.e. ecological performance; sensu Violle et al., 2007; see Table 1 for traits description). These eight traits were inflated into values ranging between 0 and 1 (a complete description of the methods is presented in Table 1). A total of thirteen trait states were used in the analyses and are hereafter referred to as "traits".

2.5. Analyses

The species diversity found in the different stand stages in peat fields and ditches was calculated with the Shannon–Weaver species diversity index (H' ; Whittaker, 1972) while the functional dispersion of the traits was estimated using the FD_{is} index (Laliberté and Legendre, 2010). As the FD_{is} index is an attribute of functional diversity, it is referred to as “functional diversity” hereafter. We used a paired t -test to compare peat fields and ditches prior to restoration and also at the different stages after restoration ($n=8$, p is significant when ≤ 0.05). Then, to ensure comparability between datasets from the experimental site and the reference ecosystem, which were surveyed using different methods, we calculated the relative contribution of each species to its community (p_i). The data table $Y=[y_{ij}]$ presents sites in rows and species occurrence (restored site) or abundance (reference ecosystem) in columns. Each species occurrence or abundance was divided by the row sums y_i (sensu Legendre and Gallagher, 2001). The community weighted mean (CWM) was calculated by multiplying the relative abundance (p_i) of each species by its inflated trait value x_i (Moretti et al., 2010) in order to obtain the mean trait score. Principal component analysis (PCA) was then applied on the CWM matrix to show associations of the herbaceous community traits to the different stand stages and the reference ecosystem. One PCA was conducted for each of the two initial drainage conditions. Equiprobability ellipses including 75% of peat fields or ditches for a given stand stage or 75% of peatlands composing the reference ecosystem studied were drawn. This visual representation enabled observation to determine whether the studied stand stages had low or high trait diversity (i.e. sites respectively grouped as small or large ellipses). The fourth-corner analysis (permutation model one) was used to select the eight significant traits (Aubin et al., 2009; Dray and Legendre, 2008) and helped in data interpretation of the CWM analysis procedure (see supplementary material, Appendix B, Table 2.a and 2.b). PCA performed on the species (p_i) matrix transformed with Hellinger was used to choose representative species for the different stand stages and the reference ecosystem. One PCA was conducted for each of the two initial drainage conditions (see supplementary material, Appendix C, Fig. 3.a and 3.b). All statistical analyses were performed using R 2.10.1 (R Development Core Team, 2009), more precisely with the FD 1.0–11 (Laliberté and Shipley, 2011), Vegan 1.17–2 (Oksanen et al., 2010), Ellipse 0.3–5 (Murdoch and Chow, 2007), and Ade4 1.4–17 (Chessel et al., 2004) packages.

3. Results

3.1. Evaluation of restoration at trait level

In peat fields and ditches, traits of the herbaceous species converged toward the reference ecosystem as time after restoration advanced (see axis I where respectively 48% and 59% of the variation was explained for peat fields and ditches; Fig. 1a and Fig. 1b). In both peat fields and ditches, dominant herbaceous species prior to restoration were associated to mesic soil (WatMes) with a pH higher than 5.7 – which is fen-like pH (High_pH), were ruderal (HabRud) or wetland species (HabWet) and had mycorrhiza (Mycorrhiza; Fig. 1a and Fig. 1b; Table 2.a and 2.b). In the early stage after restoration (years one to five) for both peat fields and ditches, all of the previously described traits remained important in the community but introduced species (Exotic) appeared (Fig. 1a and Fig. 1b; Table 2.a and 2.b). All these traits decreased in importance in the latter stage after restoration (years six to ten) and were not frequent in the reference

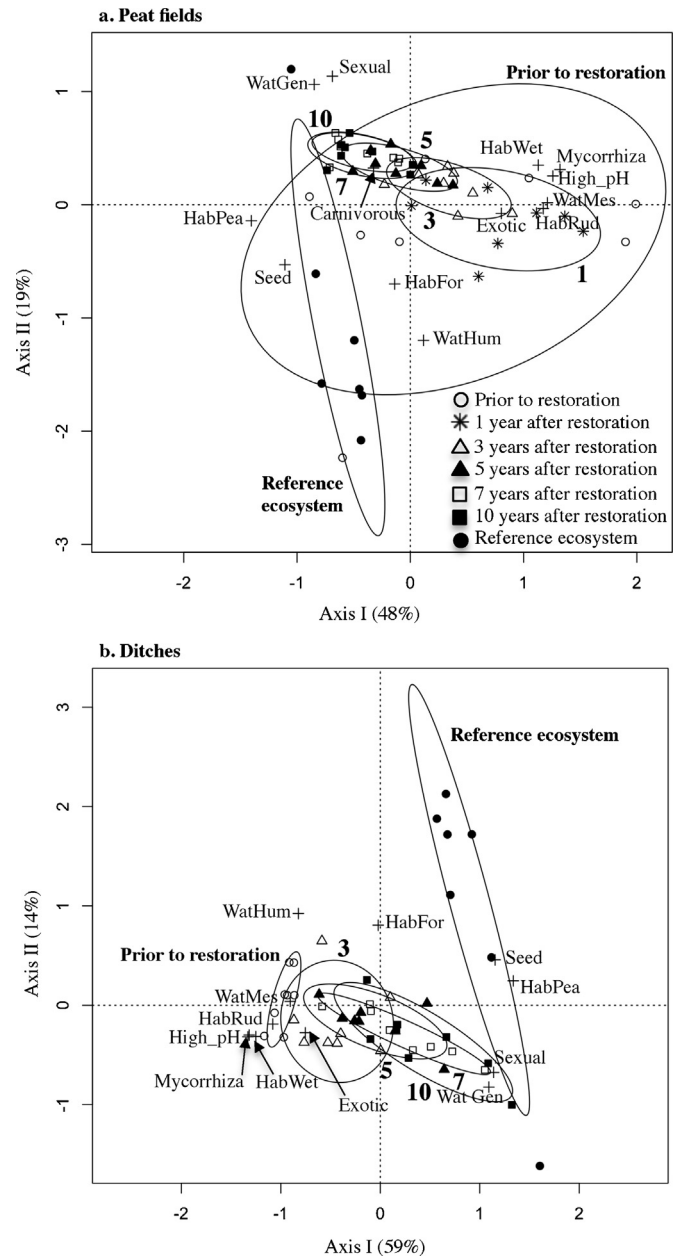


Fig. 1. Principal component analysis (PCA) diagram (axes 1 and 2) of the community weight mean matrix of the traits of the herbaceous community (see Table 1 for code) for (a) peat fields and (b) ditches as well as reference ecosystem. Equiprobability ellipses including 75% of peat fields or ditches for a given stand stage or 75% of bogs composing the reference ecosystem studied were drawn to outline a homogeneous (i.e. small ellipse) or heterogeneous (i.e. wide ellipse) set of traits of the study plots. Only the trait state with the highest inflated value is shown in the analyses (see Table 1).

ecosystem (Table 2.a and 2.b). In the restored peat fields, carnivorous species (Carnivorous), species showing mostly sexual reproduction (Sexual) and thriving in soils with variable water availability (WatGen) arose in the latter stage after restoration (Fig. 1a and Table 2.a). Only the two latter traits were present for ditches in year ten after restoration (Fig. 1b; Table 2.b). The reference ecosystem had species associated to forest (HabFor) and peatland habitats (HabPea), as well as species associated to humid soil (WatHum) and with large seeds (Seed; Fig. 1a and Fig. 1b; Table 2.a and 2.b).

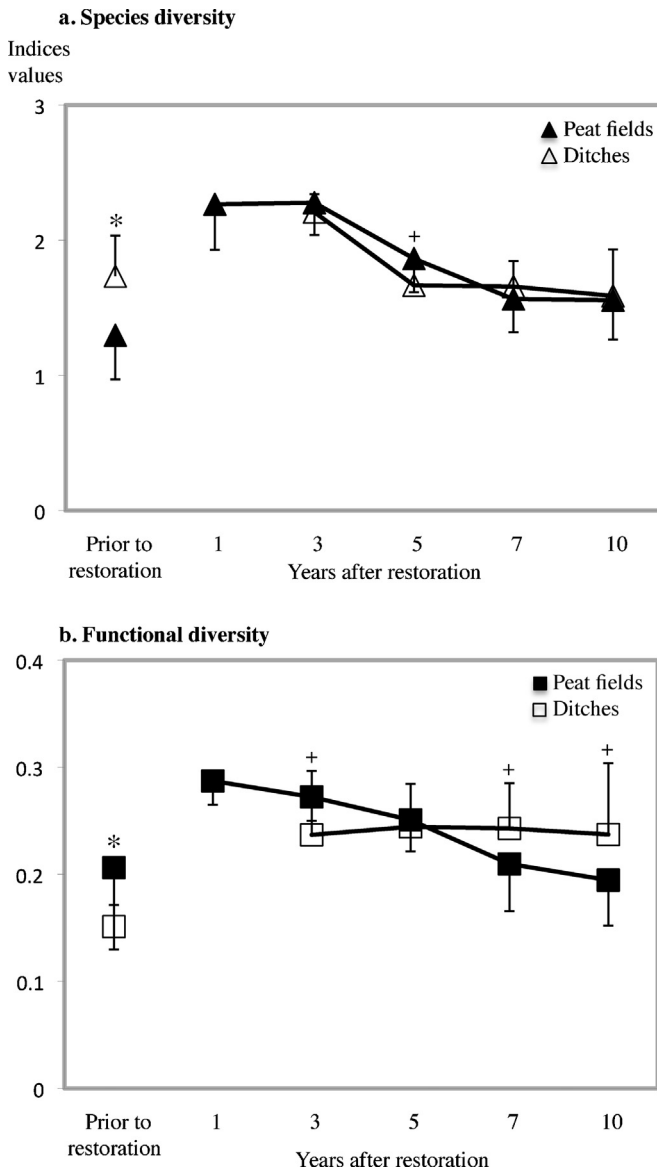


Fig. 2. Analyses showing (a) species diversity (H' – triangle) and (b) functional diversity ($FDis$ – square) of the herbaceous community in peat fields (filled triangle or square) and ditches (empty triangle or square) prior to and up to ten years after restoration. The reference ecosystem is not represented because of differences in the data collection protocol. The y-axis represents the respective index values. Standard deviation is presented by upper bars for ditches and by lower bars for peat fields. Difference between peat fields and ditches for the stage tested ($n = 8$) by the paired T -test are showed by one asterisk when $p \leq 0.05$ (significant) and a cross when $0.05 < p < 0.1$.

3.2. Response of species and functional diversity to restoration

Prior to restoration, species diversity was significantly higher in ditches (H' of 1.73 ± 0.30) than in peat fields (H' of 1.30 ± 0.33 ; Fig. 2a), while functional diversity was significantly higher in peat fields ($FDis$ of 0.21 ± 0.08 ; Fig. 2b) than in ditches ($FDis$ of 0.15 ± 0.02 ; Fig. 2b). Shortly after restoration, both indices (species and functional diversity) increased in peat fields and ditches. As time after restoration went by, species diversity (H' from 2.27 ± 0.34 to 1.56 ± 0.29 ; Fig. 2a) and functional diversity ($FDis$ from 0.28 ± 0.02 to 0.19 ± 0.04 ; Fig. 2b) decreased steadily in peat fields. In ditches, species diversity decreased (H' from 2.21 ± 0.13 to 1.59 ± 0.35 ; Fig. 2a) while functional diversity remained stable with

$FDis$ value of $\sim 0.24 \pm 0.04$ (Fig. 2b). Overall, there was no significant difference between the species diversity found in peat fields and in ditches following restoration ($p > 0.05$). On the other hand, functional diversity showed a trend where the difference between peat fields and ditches seemed to become closer to be a significant result (seven years after restoration, $p = 0.073$; ten years after restoration, $p = 0.058$).

3.3. Trait diversity found within the different stand stages and in the reference ecosystem

Across the seven undisturbed peatlands used as the reference ecosystem, traits were diversified or, at least, dominant species abundance corresponding to these traits varied among sites (see wide ellipse on axis 2 where 19% and 14% of the variation was explained for the reference ecosystem when it was compared respectively to peat fields and ditches; Fig. 1a and Fig. 1b). Similar results were found in peat fields prior to and in the early stage after restoration (i.e. wide ellipse; Fig. 1a). Ten years after restoration, trait diversity found in peat fields was low (i.e. small ellipse; Fig. 1a) while the opposite trend was seen in ditches. Indeed, species found in ditches had a highly specific set of traits prior to restoration, but as time after restoration went by, herbaceous species showed more variation in the set of traits (Fig. 1b).

4. Discussion

4.1. Is the herbaceous community of the restored peatland converging toward the reference ecosystem?

In a restoration context, the ecological distance between (1) the site prior to and after restoration and especially between (2) the restored site and the reference ecosystem is used to evaluate project success (Dufour and Piégay, 2009). In our study, for both peat fields and ditches, the herbaceous community showed similar traits composition prior to and in the early stage after restoration. However, different traits composition was identified between (1) the early and the latter stages after restoration, as well as between (2) the latter stage after restoration and the reference ecosystem. It remains that traits of the herbaceous community generally converged toward the reference ecosystem through time after restoration.

Prior to restoration, harsh conditions, few competitors and seed dispersal from surrounding fields favored the development of a herb community dominated by ruderal (HabRud) and wetland species (HabWet), associated to mesic soil conditions (WatMes) with a fen-like pH (High.pH) and with mycorrhiza. All the traits present prior to restoration remained significant in the early stage after restoration, which could in part result from one of the steps of the moss layer transfer technique when spontaneously established plants were moved into ditches during the surface leveling process. These plants may have provided seeds and rhizomes for colonizing restored surfaces once conditions were improved. However, this phenomenon had limited impact on the restoration processes as all traits important prior to and in the early stage after restoration quickly decreased in the latter stage after restoration. For example, ruderal species, which may have come from surrounding agricultural fields (Lachance and Lavoie, 2004) or disturbed areas (i.e. road side), disappeared in the latter stage after restoration. Also, herbaceous species adapted to mesic soil conditions and preferring a wetland habitat were found prior to restoration due to the presence of open water in ditches and flooding in the early stage after restoration (Shantz and Price, 2006). Both traits decreased in the herbaceous community found in the latter stage after restoration.

Finally, herbaceous species associated to a fen-like pH, with mycorrhiza and introduced species (Exotic), which are not characteristics typical of peatland species (Thormann et al., 1999b; Lavoie and Rochefort, 1996; Gorham and Janssens, 1992), were all favored in the early restoration stage. Such traits may have been favored by phosphorus application (Steers et al., 2011; Andersen et al., 2006; Jasper et al., 1979) which is recommended to facilitate the return of nurse plants for *Sphagnum* (Groeneveld and Rochefort, 2002). As herbaceous species with those traits decreased in importance within their community in the latter stage after restoration, it seems that phosphorus application does not continue to favor atypical peatland traits ten years after restoration.

Carnivorous species, rare prior to restoration, were globally well established ten years later. Carnivorous species take longer than other herbaceous species to develop their leaf structure, but eventually have access to more nutrients and energy (Karagatzides and Ellison, 2009) especially in nutrient poor habitats such as bogs. The presence of herbaceous species showing mostly sexual reproduction and linked to variable water availability (WatGen) was expected in cut-over bogs (Table 1; Pywell et al., 2003; Payette and Rochefort, 2001) but these species were dominant mainly in the latter stage after restoration, most likely due to the prominence of *E. vaginatum* (Fig. 3.a and 3.b). Several ecological characteristics explain the establishment success of *E. vaginatum*. This species produces numerous seeds easily dispersed by wind (Lavoie et al., 2005; Campbell et al., 2003). Also, it thrives in soil with variable water availability (Gignac et al., 2004) and has the ability to invade cut-over peatlands with a water table no deeper than 30–40 cm under the surface (Lavoie et al., 2005). As expected (Table 1; Pellerin and Lavoie, 2003; Payette and Rochefort, 2001; Turnbull et al., 1999), peatland (HabPea) and forest species (HabFor), species preferring humid soil (WatHum) as well as with large seeds (Seed) were associated to the reference ecosystem. Species with large seeds can be expected to be more prominent in undisturbed sites, as space and resources are limited (Turnbull et al., 1999).

4.2. Patterns of species, functional and trait diversity

Soon after restoration, species and functional diversity were at their highest in both peat fields and ditches. This is a classic example of invasion by functionally different species following the changes in the environmental filters (Mayfield et al., 2010). As time after restoration passed, species diversity decreased in both peat fields and ditches whereas functional diversity decreased in peat fields and seem to be maintained in ditches. In ditches, a strong humidity gradient was seen from their center toward adjacent peat fields, which may have favored different herbaceous composition. By its contrasting environmental conditions, the presence of ditches may help maintaining high functional diversity in restored peatlands. Nevertheless, the reference ecosystem showed higher trait diversity (see wide ellipse on Fig. 1a and Fig. 1b) than peat fields and ditches. The high trait diversity of the reference ecosystem may be an artifact of the comparison of seven undisturbed peatlands (composing the reference ecosystem) with only one restored site or could be due to natural heterogeneity usually found in “mature ecosystems” (Bartels and Chen, 2010). However, low heterogeneity in trait diversity of managed ecosystem has also been shown for forests (Bartels and Chen, 2010), grasslands (Kahmen et al., 2002), wetlands (Seabloom and van der Valk, 2003) and peatlands (D'Astous, 2012).

4.3. Recommendations for ombrotrophic peatland restoration

Experimenting with a trait-based approach as a method of assessment for a peatland restoration project has several useful

outcomes. First, our study can serve as a framework to compare ecosystems across a vast region or range of sites, as it is based on universal traits (not associated to a specific region) rather than on site-specific vegetation composition (e.g. Laliberté et al., 2010). Also, the clear changes in trait pattern with time after restoration showed the importance of monitoring a restored peatland over a period of ten years. In addition, the integration of the trait-based approach into multiple year peatland restoration monitoring can help target specific traits/process that are not self-restoring. This could improve the well-known moss layer transfer technique in a way that optimizes ecosystem recovery. For instance, it could be possible to sow species with large seeds in the latter stage after restoration in order to increase their representation within the community. Examples of such species are *Carex aquatilis* Wahlenb., *Carex oligosperma* Michx., *Eriophorum virginicum* L., and *R. chamaemorus*. A second finding of our analysis is the persistence of several traits not typical of bog species for up to five years after restoration, which may be a result of stocking residual peat in ditches during the field preparation step. Two alternative steps could prevent this phenomenon (1) the removal of spontaneously established vegetation prior to restoration, or (2) the restoration of peatlands as soon as exploitation ceases. Overall, the new steps recommended for the moss layer transfer technique, especially sowing seeds, must be evaluated in view of the goals of the restoration project and the costs they incur. It is also possible that the restored ecosystem will come to resemble the reference ecosystem more over a longer time period after restoration (such as twenty or fifty years). Finally, our study points out the relative homogeneity of a restored ecosystem compared to a reference ecosystem. If the desired goal is to bring the restored peatland closer to the reference ecosystem, diversity must be increased to foster a heterogeneous ecosystem. Our study highlights the importance of creating or preserving contrasting habitats within a restored site, as functional and trait diversity between ditches and peat fields were different. Also, new techniques to restore habitat diversity within cut-over bogs should be further developed, notably in terms of their potential application for shallow or deep ponds. We also recommend integrating vegetation material from different undisturbed bogs as part of a broader regional planning strategy to help increase diversity of restored bogs.

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Supplementary data

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Glossary

Trait (Violle et al., 2007): Traits can be any morphological, physiological or phenological characteristics of an individual that affect its growth, reproduction or survival. Traits can also be related to optimum species distribution along environmental gradients.

Functional diversity (Mason et al., 2005): Functional diversity is the distribution of the species and abundance of a community in niche space, including: (a) the amount of niche space filled by species in the community (functional richness), (b) the evenness of abundance distribution in filled niche space (functional evenness) and (c) the degree to which abundance distribution in niche space maximises divergence in functional characters within the community (functional divergence).

Cut-over peatland: In this instance, ombrotrophic peatland (bog) where industrial peat exploitation was carried out using vacuum suction.